

Operations Research, Central Exercise 2

The simplex algorithm

Janik Bürgermeister

April 24, 2026

Summary LPs

- ▶ LPs describe convex polytopes.
- ▶ There exists an optimal vertex.
- ▶ Vertices are obtained at the intersection of n constraints.
($x \in \mathbb{R}^n$)
- ▶ Intersections of n constraints are not always feasible vertices.
- ▶ Vertices are basic feasible solutions (BFSs) of the associated linear system of equations in standard form.
- ▶ Optimality can be determined by looking at adjacent BFSs.
- ▶ The simplex algorithm finds adjacent BFSs

Simplex Algorithm, Overview

1. Transform LP in Standard Form
2. Transform LP in Canonical Form
 - 2.1 2-phase method
 - 2.2 Big-M method
3. Investigate boundedness, feasibility and degeneracy.
(→ Tutorial Exercises)
4. Simplex iteration

Comparison of Big M Method and Two-Phase Method

How to find a feasible solution?

Aspect	Big M Method	Two-Phase Method
Approach	Introduce artificial variables with a large penalty (M) in the objective function.	Phase 1 minimizes the sum of artificial variables; Phase 2 solves the original problem.
Artificial Variables	Penalized directly in the objective function.	Eliminated in Phase 1 if feasible solution exists.
Advantages Disadvantages	Numerically unstable if M is large. No clear separation of feasibility and optimization.	Numerically stable. Separates feasibility from optimization.

Forms of LPs

Normal Form of LP:

$$\begin{array}{ll}\max & c^T x \\ \text{s.t.} & Ax \leq b \\ & x \geq 0\end{array}$$

Standard Form of LP:

$$\begin{array}{ll}\max & c^T x \\ \text{s.t.} & Ax = b \quad (b \geq 0) \\ & x \geq 0\end{array}$$

Canonical Form of LP:

Standard form and each (non-trivial) constraint has a variable which is normed to 1 and does not appear in all other (non-trivial) constraints.

Forms of LPs

Example for Transformation into Canonical Form of LP and graphical interpretation of the simplex algorithm:

$$\max 2x_1 + x_2$$

$$s.t. x_1 + x_2 \leq 1$$

$$x_i \geq 0 \quad \forall i \in [2]$$

$$\max 2x_1 + x_2$$

$$s.t. x_1 + x_2 + x_3 = 1$$

$$x_i \geq 0 \quad \forall i \in [3]$$

Forms of LPs

Example for Transformation into Canonical Form of LP and graphical interpretation of the simplex algorithm:

$$\max 2x_1 + x_2$$

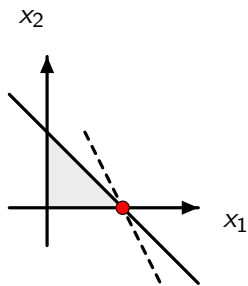
$$s.t. x_1 + x_2 \leq 1$$

$$x_i \geq 0 \quad \forall i \in [2]$$

$$\max 2x_1 + x_2$$

$$s.t. x_1 + x_2 + x_3 = 1$$

$$x_i \geq 0 \quad \forall i \in [3]$$



Example for Transformation into Canonical Form of LP and graphical interpretation of the simplex algorithm:

$$\max 2x_1 + x_2$$

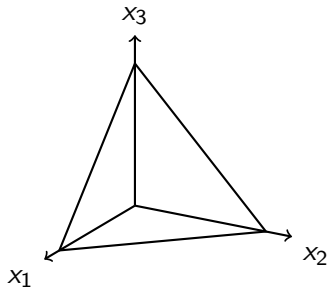
$$s.t. x_1 + x_2 \leq 1$$

$$x_i \geq 0 \quad \forall i \in [2]$$

$$\max 2x_1 + x_2$$

$$s.t. x_1 + x_2 + x_3 = 1$$

$$x_i \geq 0 \quad \forall i \in [3]$$



Exercise 1

A storage room is to be equipped with shelves. The room has a total of $72m^2$ of floor space available for shelves.

There are two types of shelves to choose from. The first shelf costs 10€ , requires $6m^2$ of floor space, and provides $8m^3$ of storage capacity. The second shelf costs 20€ , requires $8m^2$ of floor space, and provides $12m^3$ of storage capacity.

How many shelves of each type would you buy to maximize the storage capacity if your budget is 140€ ? Determine the optimal solution using the Simplex algorithm.

Exercise 1

A storage room is to be equipped with shelves. The room has a total of $72m^2$ of floor space available for shelves.

There are two types of shelves to choose from. The first shelf costs 10€ , requires $6m^2$ of floor space, and provides $8m^3$ of storage capacity. The second shelf costs 20€ , requires $8m^2$ of floor space, and provides $12m^3$ of storage capacity.

How many shelves of each type would you buy to maximize the storage capacity if your budget is 140€ ? Determine the optimal solution using the Simplex algorithm.

Shelf	Costs	Floor Space	Capacity
1			
2			
Total			

Exercise 1, Modelling

Shelf	Costs	Floor Space	Capacity
1	10	6	8
2	20	8	12
Total	140	72	max

x_1 Number of shelves of the 1st type purchased

x_2 Number of shelves of the 2nd type purchased

Write down the corresponding LP and convert it into canonical form. What is the initial basis? (2 min)

Exercise 1, Modelling

Shelf	Costs	Floor Space	Capacity
1	10	6	8
2	20	8	12
Total	140	72	max

x_1 Number of shelves of the 1st type purchased

x_2 Number of shelves of the 2nd type purchased

Write down the corresponding LP and convert it into canonical form. What is the initial basis? (2 min)

$$\begin{aligned} \max \quad & 8x_1 + 12x_2 \\ \text{s.t.} \quad & 10x_1 + 20x_2 \leq 140 \\ & 6x_1 + 8x_2 \leq 72 \\ & x_i \geq 0 \quad i \in [2] \end{aligned}$$

Exercise 1, Conversion to Standard form

Normal Form

$$\begin{aligned} \max \quad & 8x_1 + 12x_2 \\ \text{s.t.} \quad & 10x_1 + 20x_2 \leq 140 \\ & 6x_1 + 8x_2 \leq 72 \\ & x_i \geq 0 \quad i \in [2] \end{aligned}$$

Standard Form

Exercise 1, Conversion to Standard form

Normal Form

$$\begin{aligned} \max \quad & 8x_1 + 12x_2 \\ \text{s.t.} \quad & 10x_1 + 20x_2 \leq 140 \\ & 6x_1 + 8x_2 \leq 72 \\ & x_i \geq 0 \quad i \in [2] \end{aligned}$$

Standard Form

$$\begin{aligned} \max \quad & 8x_1 + 12x_2 = z \\ \text{s.t.} \quad & 10x_1 + 20x_2 + x_3 = 140 \\ & 6x_1 + 8x_2 + x_4 = 72 \\ & x_i \geq 0 \quad i \in [4] \end{aligned}$$

Exercise 1, Simplex

Initial basis: x_3, x_4 (LP is already in canonical form)

$$\max z$$

$$\text{s.t. } z - 8x_1 - 12x_2 = 0$$

$$10x_1 + 20x_2 + x_3 = 140$$

$$6x_1 + 8x_2 + x_4 = 72$$

$$x_i \geq 0 \quad i \in [4]$$

Exercise 1, Simplex

Initial basis: x_3, x_4 (LP is already in canonical form)

max z

$$\text{s.t. } z - 8x_1 - 12x_2 = 0$$

$$10x_1 + 20x_2 + x_3 = 140$$

$$6x_1 + 8x_2 + x_4 = 72$$

$$x_i \geq 0 \quad i \in [4]$$

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

Exercise 1, Simplex

Simplex Iteration

1. Find pivot column j : (most) negative entry in z-row.
This is the new basic variable.
2. Find pivot row i : $\arg \min \left\{ \frac{b_i}{a_{i,j}}, a_{i,j} > 0 \right\}$.
This is the outgoing basic variable.
3. Bring Tableau in canonical form with respect to the new basis. Only use row operations.

Exercise 1, Simplex

1. Find pivot column j : most negative entry in z -row.
This is the new basic variable.

2. Find pivot row i :
 $\arg \min \left\{ \frac{b_i}{a_{i,j}}, a_{i,j} > 0 \right\}$.
This is the outgoing basic variable.

(1 min)

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

Exercise 1, Simplex

1. Find pivot column j : most negative entry in z -row.
This is the new basic variable.

2. Find pivot row i :
 $\arg \min \left\{ \frac{b_i}{a_{i,j}}, a_{i,j} > 0 \right\}$.
This is the outgoing basic variable.

(1 min)

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

New basic variable is x_2

Exercise 1, Simplex

1. Find pivot column j : most negative entry in z -row.
This is the new basic variable.

2. Find pivot row i :
 $\arg \min \left\{ \frac{b_i}{a_{i,j}}, a_{i,j} > 0 \right\}$.
This is the outgoing basic variable.

(1 min)

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

New basic variable is x_2

$$\frac{140}{20} = 7 < 9 = \frac{72}{8}$$

Exercise 1, Simplex

1. Find pivot column j : most negative entry in z -row.
This is the new basic variable.

2. Find pivot row i :
 $\arg \min \left\{ \frac{b_i}{a_{i,j}}, a_{i,j} > 0 \right\}$.
This is the outgoing basic variable.

(1 min)

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

New basic variable is x_2

$$\frac{140}{20} = 7 < 9 = \frac{72}{8}$$

Outgoing basic variable is x_3

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

Exercise 1, Simplex

Simplex Iteration

3. Bring Tableau in canonical form with respect to the new basis $\{x_2, x_4\}$. Only use row operations.

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z						
x_2						
x_4						

Exercise 1, Simplex

Simplex Iteration

3. Bring Tableau in canonical form with respect to the new basis $\{x_2, x_4\}$. Only use row operations.

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-8	-12			0
x_3		10	20	1		140
x_4		6	8		1	72

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2	0	$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2	0	$-2/5$	1	16

Exercise 1, Simplex

Perform one iteration of the simplex algorithm (5 min)

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2		$-2/5$	1	16

Exercise 1, Simplex

1. New basic variable is x_1 :

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2		$-2/5$	1	16

Exercise 1, Simplex

1. New basic variable is x_1 :

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2		$-2/5$	1	16

2. $\frac{7}{\frac{1}{2}} = 14 > 8 = \frac{16}{2}$

Exercise 1, Simplex

1. New basic variable is x_1 :

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		3/5		84
x_2		1/2	1	1/20		7
x_4		2		-2/5	1	16

$$2. \frac{7}{\frac{1}{2}} = 14 > 8 = \frac{16}{2}$$

Outgoing basic variable is x_4 :

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		3/5		84
x_2		1/2	1	1/20		7
x_4		2		-2/5	1	16

Exercise 1, Simplex

Simplex Iteration

3. Bring Tableau in canonical form with respect to the new basis $\{x_1, x_2\}$. Only use row operations.

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2		$-2/5$	1	16

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z		0				
x_2		0				
x_1		1				

Exercise 1, Simplex

Simplex Iteration

3. Bring Tableau in canonical form with respect to the new basis $\{x_1, x_2\}$. Only use row operations.

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2		$-2/5$	1	16

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	0		$1/5$	1	100
x_2		0	1	$3/20$	$-1/4$	3
x_1		1		$-1/5$	$1/2$	8

Exercise 1, Simplex

Simplex Iteration

3. Bring Tableau in canonical form with respect to the new basis $\{x_1, x_2\}$. Only use row operations.

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		2		$-2/5$	1	16

Base	Coefficients					Res.
	z	x_1	x_2	x_3	x_4	
z	1	0		$1/5$	1	100
x_2		0	1	$3/20$	$-1/4$	3
x_1		1		$-1/5$	$1/2$	8

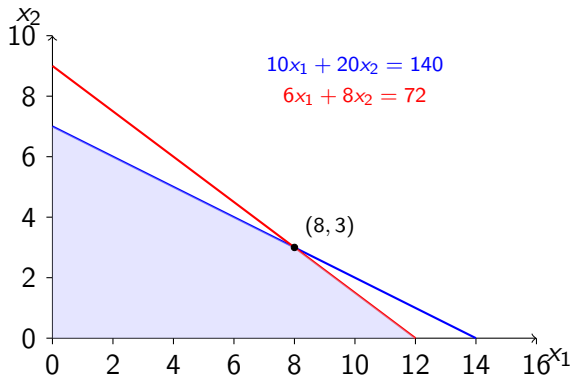
Termination: only non-negative values in the z-row.

Optimal solution: $x_1^* = 8$, $x_2^* = 3$, with objective value $z^* = 100$.

Exercise 1, Sanity Check, Graphical solution

Simplex solution: $x_1^* = 8, x_2^* = 3$, with objective value $z^* = 100$.

$$\begin{aligned} \max \quad & 8x_1 + 12x_2 \\ \text{s.t.} \quad & 10x_1 + 20x_2 \leq 140 \\ & 6x_1 + 8x_2 \leq 72 \\ & x_1, x_2 \geq 0 \end{aligned}$$



Exercise 2

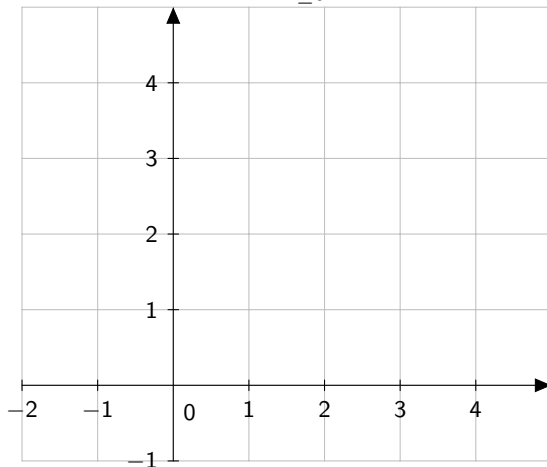
Consider the following Linear Program (LP):

$$\begin{array}{ll} \text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{array}$$

1. Solve the LP graphically.
2. Transform the LP to standard form.
3. Is the LP in canonical form? Justify. (5 min)
4. Compute an optimal basic solution for the LP using the Simplex algorithm, utilizing the Big-M method.
5. Mark the feasible basic solutions found in part (d) on your graphical solution from part (a).

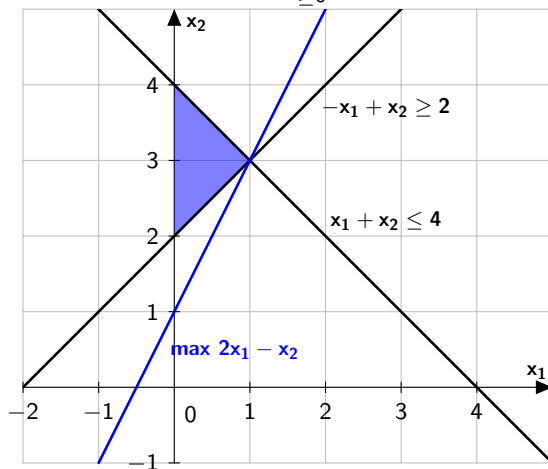
Exercise 2, a) Graphical Solution

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2\end{array}$$



Exercise 2, a) Graphical Solution

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2\end{array}$$



Exercise 2, b) and c)

Transform to standard form

$$\begin{array}{ll} \text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{array}$$

Exercise 2, b) and c)

Transform to standard form

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2\end{array}$$

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 - x_3 = 2 \\ & x_1 + x_2 + x_4 = 4 \\ & x \in \mathbb{R}_{\geq 0}^4\end{array}$$

Exercise 2, b) and c)

Transform to standard form

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2\end{array}$$

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 - x_3 = 2 \\ & x_1 + x_2 + x_4 = 4 \\ & x \in \mathbb{R}_{\geq 0}^4\end{array}$$

The LP is not in canonical form since the first constraint lacks a basic variable.

Exercise 2, d)

We need an additional artificial variable $w_1 \geq 0$ for the first constraint.

$$\begin{array}{ll} \text{Max} & 2x_1 - x_2 - Mw_1 = z \\ \text{s.t.} & -x_1 + x_2 - x_3 + w_1 = 2 \\ & x_1 + x_2 + x_4 = 4 \\ & x \in \mathbb{R}_{\geq 0}^4 \\ & w_1 \geq 0 \end{array}$$

Exercise 2, d)

We need an additional artificial variable $w_1 \geq 0$ for the first constraint.

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 - Mw_1 = z \\ \text{s.t.} & -x_1 + x_2 - x_3 + w_1 = 2 \\ & x_1 + x_2 + x_4 = 4 \\ & x \in \mathbb{R}_{\geq 0}^4 \\ & w_1 \geq 0\end{array}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-2	1			M	0
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Exercise 2, d)

The tableau needs to be transformed into canonical form first. For this purpose, variable w_1 must be eliminated from the objective. (1 min)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-2	1			M	0
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Exercise 2, d)

The tableau needs to be transformed into canonical form first. For this purpose, variable w_1 must be eliminated from the objective. (1 min)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-2	1			M	0
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	$-2 + M$	$1 - M$	M			$-2M$
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Exercise 2, d)

Perform an iteration of the simplex algorithm (5 min)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	$-2 + M$	$1 - M$	M			$-2M$
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Basic variable exchange?

Exercise 2, d)

Perform an iteration of the simplex algorithm (5 min)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	$-2 + M$	$1 - M$	M			$-2M$
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Basic variable exchange? $w_1 \leftrightarrow x_2$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z		0				
x_2		1				
x_4		0				

Exercise 2, d)

Perform an iteration of the simplex algorithm

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	$-2 + M$	$1 - M$	M			$-2M$
w_1	-1	1	-1		1	2
x_4	1	1		1		4

Basic variable exchange: $w_1 \leftrightarrow x_2$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-1		1		$-1 + M$	-2
x_2	-1	1	-1		1	2
x_4	2		1	1	-1	2

Optimal?

Exercise 2, d)

Perform another iteration of the simplex algorithm, since stopping criterion is not yet met.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-1		1		$-1 + M$	-2
x_2	-1	1	-1		1	2
x_4	2		1	1	-1	2

Basic variable exchange?

Exercise 2, d)

Perform another iteration of the simplex algorithm, since stopping criterion is not yet met.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-1		1		$-1 + M$	-2
x_2	-1	1	-1		1	2
x_4	2		1	1	-1	2

Basic variable exchange? $x_4 \leftrightarrow x_1$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	0	0				
x_2	0	1				
x_1	1	0				

Optimal?

Exercise 2, d)

Perform another iteration of the simplex algorithm, since stopping criterion is not yet met.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-1		1		$-1 + M$	-2
x_2	-1	1	-1		1	2
x_4	2		1	1	-1	2

Basic variable

exchange: $x_4 \leftrightarrow x_1$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z			1.5	0.5	$-1.5 + M$	-1
x_2		1	-0.5	0.5	0.5	3
x_1	1		0.5	0.5	-0.5	1

The stopping criterion is fulfilled (only non-negative entries in the objective row). The optimal solution is $x^* = (1, 3)$ with an associated objective value $z_{\max} = -1$.

Exercise 2, e)

Mark the feasible basic solutions found in part (d) on your graphical solution from part (a).

Bases found: $\{x_2, x_4\}$ and $\{x_1, x_2\}$.

$$-x_1 + x_2 - x_3 + w_1 = 2$$

$$x_1 + x_2 + x_4 = 4$$

Exercise 2, e)

Mark the feasible basic solutions found in part (d) on your graphical solution from part (a).

Bases found: $\{x_2, x_4\}$ and $\{x_1, x_2\}$.

$$-x_1 + x_2 - x_3 + w_1 = 2$$

$$x_1 + x_2 + x_4 = 4$$

$\{x_2, x_4\}$:

$$x_2 = 2$$

$$x_2 + x_4 = 4$$

$$z_1 = (0, 2)$$

Exercise 2, e)

Mark the feasible basic solutions found in part (d) on your graphical solution from part (a).

Bases found: $\{x_2, x_4\}$ and $\{x_1, x_2\}$.

$$-x_1 + x_2 - x_3 + w_1 = 2$$

$$x_1 + x_2 + x_4 = 4$$

$\{x_2, x_4\}$:

$$x_2 = 2$$

$$x_2 + x_4 = 4$$

$$z_1 = (0, 2)$$

$\{x_1, x_2\}$:

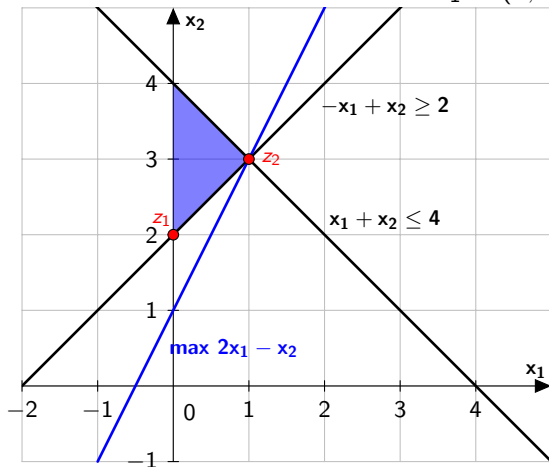
$$-x_1 + x_2 = 2$$

$$x_1 + x_2 = 4$$

$$z_2 = (1, 3)$$

Exercise 2, e)

The two feasible basic solutions are $z_1 = (0, 2)$ and $z_2 = (1, 3)$.



Exercise 3, How to handle unbounded variables.

Solve the following linear optimization problem using the Simplex algorithm:

Max

$$4x_1 + x_2 = z$$

s.t.

$$x_1 + x_2 \leq 1$$

$$x_2 \leq 1$$

$$x_1, x_2 \in \mathbb{R}$$

Exercise 3, Transformation

The variables x_1 and x_2 are unbounded. However, to still be able to apply the Simplex algorithm, x_1 is replaced by $x_1^+ - x_1^-$ and x_2 by $x_2^+ - x_2^-$, where: $x_1^+, x_1^-, x_2^+, x_2^- \geq 0$.

Max

$$4x_1^+ - 4x_1^- + x_2^+ - x_2^- = z$$

s.t.

$$x_1^+ - x_1^- + x_2^+ - x_2^- \leq 1$$

$$x_2^+ - x_2^- \leq 1$$

$$x_1^+, x_1^-, x_2^+, x_2^- \geq 0$$

Exercise 3

Max

$$4x_1^+ - 4x_1^- + x_2^+ - x_2^- = z$$

s.t.

Standard Form:

$$x_1^+ - x_1^- + x_2^+ - x_2^- + x_3 = 1$$

$$x_2^+ - x_2^- + x_4 = 1$$

$$x_1^+, x_1^-, x_2^+, x_2^-, x_3, x_4 \geq 0$$

Exercise 3

Max

$$4x_1^+ - 4x_1^- + x_2^+ - x_2^- = z$$

s.t.

Standard Form:

$$x_1^+ - x_1^- + x_2^+ - x_2^- + x_3 = 1$$

$$x_2^+ - x_2^- + x_4 = 1$$

$$x_1^+, x_1^-, x_2^+, x_2^-, x_3, x_4 \geq 0$$

Initial Basis x_3, x_4

Base	Coefficients							Res.
	z	x_1^+	x_1^-	x_2^+	x_2^-	x_3	x_4	
z	1	-4	4	-1	1			0
x_3	0	1	-1	1	-1	1	0	1
x_4		0		1	-1		1	1

Exercise 3

Base	Coefficients							Res.
	z	x_1^+	x_1^-	x_2^+	x_2^-	x_3	x_4	
z	1	-4	4	-1	1			0
x_3	0	1	-1	1	-1	1	0	1
x_4		0		1	-1		1	1

Basis Exchange: $x_3 \leftrightarrow x_1^+$

Exercise 3

Base	Coefficients							Res.
	z	x_1^+	x_1^-	x_2^+	x_2^-	x_3	x_4	
z	1	-4	4	-1	1			0
x_3	0	1	-1	1	-1	1	0	1
x_4		0		1	-1		1	1

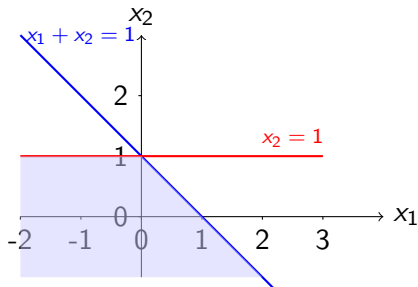
Basis Exchange: $x_3 \leftrightarrow x_1^+$

Base	Coefficients							Res.
	z	x_1^+	x_1^-	x_2^+	x_2^-	x_3	x_4	
z	1			3	-3	4		4
x_1^+		1	-1	1	-1	1		1
x_4				1	-1		1	1

The objective function coefficient at x_2^- is $-3 < 0$, and all other coefficients in the x_2^- column are negative. Thus, the objective function value is unbounded, and the algorithm terminates.

Unbounded feasible region does not imply unbounded objective value

$$\begin{array}{ll}\max & x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 \leq 1 \\ & x_2 \leq 1 \\ & x_1, x_2 \in \mathbb{R}\end{array}$$



Bonus: How to handle non-convex polytopes

Bonus: Which paths can the Simplex algorithm take?

Notes